

Technical Note

Subject - Effective Service Life of BaseLok GeoCell & GeoGrid

“What is the service life?” This is a question that we get at Industrial Fabrics, Inc. on a nearly daily basis. My typical answer is that the BaseLok products are inert and once buried or covered will be in place indefinitely. While that answer is a quick response that is true for all practical applications, the full answer is a bit more complicated. The goal of this paper is to provide a more complete answer to this ever-present question surrounding geosynthetics today.

There are many materials utilized in geosynthetics. These materials include High Density Polyethylene (HDPE), Polypropylene (PP), Polyester (PET), Linear Low Density Polyethylene (LLDPE), and Polyvinyl Chloride (PVC) to name a few. These products commonly consist of the pure resin, carbon black (UV protection), additives (antioxidants), and at times a plasticizer (PVC) and fillers. BaseLok GeoCell is made with HPDE and BaseLok GeoGrid is comprised of PP. When considering these BaseLok products, long term degradation is primarily a function of temperature, exposure to ultraviolet light, oxidation, and chemical exposure. Other factors can also exist including ozone, biological factors, hydrolysis, and radioactivity; however, these additional factors are typically negligible unless specifically present at an installation site.

In GRI White Paper #6 on Geomembrane Lifetime Prediction: Unexposed and Exposed Conditions, Robert M. Koerner, Grace Hsuan, and George R. Koerner explain that there are 3 primary stages to degradation of geomembranes in unexposed applications – Stages A, B, and C. Stage A is Antioxidant Depletion Time. Antioxidants are sometimes used to avoid polymer degradation during processing and keep oxidation from occurring during Stage A of the service life. Stage B is Induction Time to the Onset of Degradation. In a pure polyolefin (HDPE or PP) resin, pure meaning without carbon black or antioxidants, oxidation happens slowly initially, then accelerates, then decelerates. The initial slow period is Stage B and is called the induction period. Stage C is the Time to Reach 50% Degradation also known as the half-life. As oxidation begins after Stages A and B, the material begins to degrade. Often, a specific design property will be used to define this degradation. In other words, if tensile strength is used for determination of the “half-life”, then Stage C is the time it takes to reach 50% of the tensile strength. It should be noted that very little degradation takes place during Stages A and B.

Since HPDE Geomembranes are widely utilized in environmental applications, there is ample research on the longevity of HDPE. Since BaseLok GeoCell is manufactured from strips of HDPE Geomembrane, this data is directly applicable to BaseLok GeoCell. This research shows that the primary determining factor in longevity is In-Service Temperature. See the table below from the previously mentioned white paper.

Table 2 - Lifetime prediction of HDPE (nonexposed) at various field temperatures

In Service Temperature (°C)	Stage “A” (years)			Stage “B” (years)	Stage “C” (years)	Total Prediction* (years)
	Standard OIT	High Press. OIT	Average OIT			
20	200	215	208	30	208	446
25	135	144	140	25	100	265
30	95	98	97	20	49	166
35	65	67	66	15	25	106
40	45	47	46	10	13	69

*Total = Stage A (average) + Stage B + Stage C

When considering the effective design life of BaseLok GeoCell, we typically use an in-service average temperature of 25°C (77°F). The BaseLok GeoCell is buried, and it is not consistently exposed to direct sunlight. Additionally, to be conservative, we do not consider the effect of antioxidants and therefore do not consider Stage A. This results in an effective design life based on the above table from GRI White Paper #6 of 125 years. When considering any added antioxidants, the life increases to 265 years. 125 years is well beyond the typical design life of any geocell reinforced application, and that life is only applicable when the BaseLok GeoCell would be exposed to something that would degrade the HDPE.

There is far less data on the life expectancy of PP because PP is not used to make geomembranes. However, as previously mentioned, both HPDE and PP are in the polyolefin family and both have similar characteristics. PP has a higher heat resistance and given that temperature so greatly affects the life of HPDE, it stands to reason that PP products would have a longer effective design life than their HDPE sisters. Therefore, it is safe to assume an effective design life of BaseLok GeoGrid (PP) equal to that of the BaseLok GeoCell (HDPE) – 125 years.

We hope this paper provides some clarity as to the longevity of our BaseLok GeoCell and GeoGrid product.

References:

Koerner, R.M., Hsuan, G.Y. & Koerner, G.R., GRI White Paper #6 “Geomembrane Lifetime Prediction: Unexposed and Exposed Conditions.” *Geosynthetic Institute* (June 7, 2005)